Basic problems in combinatorics

Source: bijective proofs by Richard Stanley

1. The number of subsets of an n-element set is 2^n.
2. The number of even-sized and odd-sized subsets of a set are equal.
3. Let X be a set of n elements. The number of ways to choose an ordered pair (A, B) of two disjoint subsets of X is 3^n. What is the number of ways to choose (A, B) such that $A \subseteq B \subseteq X$?
4. $2n$ players are participating in a tennis tournament. Find the number P_n of pairings for the first round.
5. A composition of n is an ordered sequence $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_k)$ of positive integers such that: $\sum_{i=1}^{k} \alpha_i = n$. The number of compositions of n is given by: 2^{n-1}.
6. The total number of *parts* of all compositions of n is given by the formula: $(n+1) \cdot 2^{n-2}$.
7. For $n \geq 2$, the number of compositions of n with an even number of even parts is equal to 2^{n-2}.
8. A *multiset* M is, informally, a set with repeated elements, such as $\{1, 1, 1, 2, 4, 4, 4, 5, 5\}$, abbreviated $\{1^3, 2, 4^3, 5^2\}$ The multiplicity of element i is denoted by $\nu(i)$. The number of appearances of each element is given by $\nu(i)$. The definition of a submultiset N of M should be clear, viz.,

   $$\nu_N(i) \leq \nu_M(i) \text{ for all } i.$$

   Let $M = \{1^{\nu_1}, 2^{\nu_2}, \ldots, k^{\nu_k}\}$. How many submultisets does M have?
9. The size or cardinality of a multiset M, denoted $\#M$ or $|M|$, is its number of elements, counting repetitions. For instance, if

   $$M = \{1, 1, 1, 2, 4, 4, 4, 5, 5\}$$

   then $\#M = 9$.

   A multiset M is said to be on a set S if every element of M is an element of S. Let N_k denote the number of k-element multisets on an n-set, i.e., the number of ways of choosing, without regard to order, k elements from an n-element set if repetitions are allowed. Then, the formula for the number of k-element multisets is given by: $N_k = \binom{n+k-1}{k}$.
10. Fix $k, n \geq 0$ Find the number of solutions in nonnegative integers to $x_1 + x_2 + \cdots + x_k = n$.
11. Number of monomials in $x_1, \ldots, x_n$ of degree at most d is $\binom{n+d}{d}$
12. Fix positive integers n and k. We aim to find the number of k-tuples $(S_1, S_2, \ldots, S_k)$ of subsets S_i of $\{1, 2, \ldots, n\}$ under the following conditions:

    (a) $S_1 \subseteq S_2 \subseteq \cdots \subseteq S_k$

    (b) The sets $S_1, S_2, \ldots, S_k$ are pairwise disjoint.

(c) $S_1 \cap S_2 \cap \cdots \cap S_k = \emptyset$

13. The Fibonacci numbers F_n are defined by $F_1 = F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 2$. The number $f(n)$ of compositions of n with parts 1 and 2 is F_{n+1}.

14. The number of compositions of n with all parts > 1 is F_{n-1}.

15. The number of compositions of n with odd parts is F_n.

16. How many subsets S of $\{1, 2, \ldots, n\}$ don't contain two consecutive integers?

17. How many binary sequences (i.e., sequences of 0's and 1's) $(\epsilon_1, \ldots \epsilon_n)$ satisfy $\epsilon_1 \leq \epsilon_2 \geq \epsilon_3 \leq \epsilon_4 \geq \epsilon_5 \leq \ldots$?

18. Let $m, n \geq 0$. How many lattice paths are there from $(0, 0)$ to (m, n), if each step in the path is either $(1, 0)$ or $(0, 1)$?

19. Give bijective proof: $2\binom{2n-1}{n} = \binom{2n}{n}$.

20. Give bijective proof: For $n \geq 1$:
$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$$

21. Give bijective proof: For $n \geq 0$:
$$\sum_{k=0}^{n} \binom{x}{k}\binom{y}{n-k} = \binom{x+y}{n}$$

22. For $n \geq 0$:
$$\sum_{k=0}^{n} \binom{x+k}{k} = \binom{x+n+1}{n}$$

23. How many $m \times n$ matrices of 0's and 1's are there, such that every row and column contains an odd number of 1's?

24. Fix $k, n \geq 1$. The number of sequences $a_1, a_2, \ldots, a_n$ such that
$$1 \leq a_i \leq k \quad \text{and} \quad a_i \neq a_{i+1} \quad \text{for } 1 \leq i < n$$
is given by: $k(k-1)^{n-1}$

25. Fix positive integers n and k. Find the number of k-tuples $(S_1, S_2, \ldots, S_k)$ of subsets S_i of $\{1, 2, \ldots, n\}$ satisfying $S_1 \subseteq S_2 \supseteq S_3 \subseteq S_4 \supseteq S_5 \subseteq \cdots$

26. A binary word of length n is a sequence $(w_1, w_2, \ldots, w_n)$ where $w_i \in \{0, 1\}$ for all $i \in [n]$. They are usually written without commas and brackets: 1101001 is a binary word of length 7. Let w be any binary word of length k. Given $n \geq k$, count the binary words of length n that do not contain w as a subsequence. *Hint:* This number only depends on n and k.